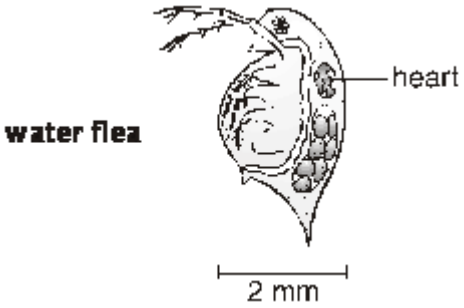
Q1.

Kava is a drug. It dissolves in alcohol but **not** in water.
A scientist tested kava to see if it can reduce the human heart rate.
Before testing the drug on humans, she tested it on water fleas.



(a) She gave two groups of water fleas a different treatment.

group	number of water fleas	treatment
1	20	one drop of kava dissolved in alcohol
2	20	one drop of alcohol

- She placed the water fleas in a dish of water under a microscope.
- She measured the heart rate of each water flea before the treatment.
- She waited 30 seconds after the treatment was given and measured the heart rate again.
- She calculated the average heart rate for each group.

(i) Why did the scientist measure the heart rate of the water fleas before the treatment?

.....

.....

1 mark

(ii) After giving the treatment, why did she wait for 30 seconds before measuring the heart rate?

.....

.....

1 mark

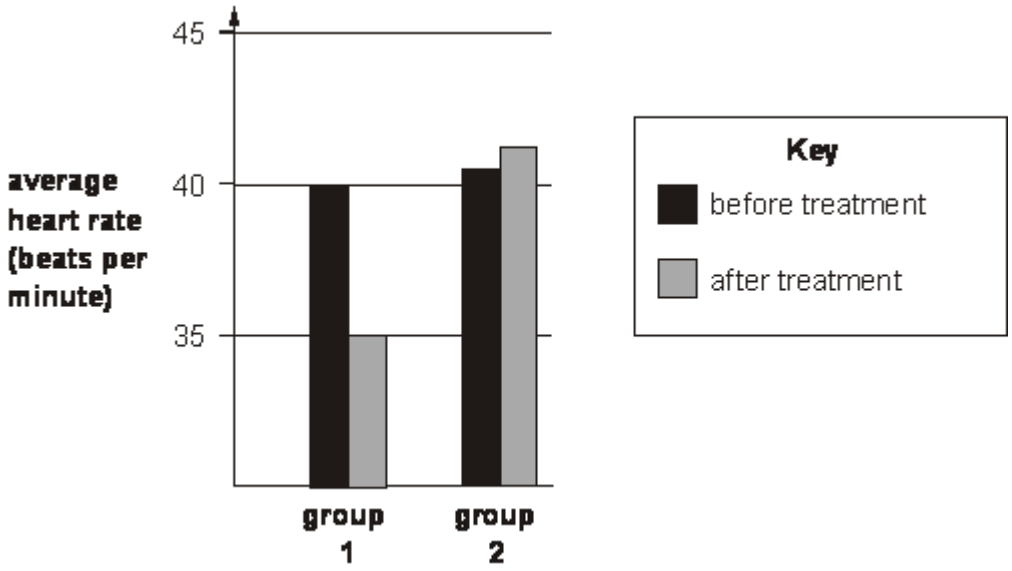
(iii) How could the scientist increase the reliability of the investigation?

.....

.....

1 mark

(b) The results of the experiment are shown below.



(i) How will the results from group 2 help in the experiment?

.....

.....

1 mark

(ii) How can the scientist use the results above to work out the effect of kava **alone** on the average heart rate of water fleas?

.....

.....

1 mark

(c) From this experiment, why could she **not** be certain how kava will affect humans?

.....

.....

1 mark
maximum 6 marks

Q1.

(a) (i) any **one** from

- to see how the heart rate had changed
  accept 'to see the result of the treatment'
  or *'to see if the treatment made a difference'*
  'to make it a fair test' ***or*** *'for a control'*
  are insufficient responses

- to compare the rate before and after the treatment
  accept 'to get a baseline ***or*** *standard'*
  or *'to make a comparison'*

1 (L7)

(ii) • so the water fleas could adjust to the solution **or** settle down
accept 'to allow time for the drug to work'
accept 'to allow the treatment to get around its body'

1 (L7)

(iii) • use more than 20 water fleas in each group
accept 'use more water fleas'
accept 'repeat it with the same fleas after the effects have worn off'
'repeat it' is insufficient

1 (L7)

(b) (i) • to show if alcohol **or** kava affects the heart rates **or** water fleas
accept 'it shows alcohol alone increases ***or*** *affects the heart rate'*
'as a comparison' is insufficient
accept 'to see what effect alcohol has'
accept 'to show it is the kava that affects ***or*** *reduces the heart rate of fleas'*
accept 'to show it is not alcohol that affects ***or*** *reduces the heart rate'*
accept '(it acts) as a control'

1 (L7)

(ii) any **one** from

- take the average heart rate resulting from kava in alcohol away from the average heart rate resulting from alcohol
  accept 'add the increase with alcohol to the decrease with kava and alcohol'

- take the difference between kava in alcohol and alcohol only
  accept 'subtract 35 from 41 ***or*** *42'*

- look at the difference between group 1 and group 2 after treatment 1

1 (L7)

(c) any **one** from

- there may be side effects
- the effect could be different in humans
- it has not been tested on humans
- human hearts are different from fleas' hearts

*accept 'humans are bigger **or** from a different group **or** species'*

*'they are not the same' **or** 'they are different animals' are insufficient responses*

*accept 'we do not know anything about the dose **or** amount given'*

1 (L7)

[6]